Arista Access Node BTM-ACC-19 Optical Transport Failure SOP

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_2640

Description

Optical Transport Failure on Arista Access Node BTM-ACC-19 in the BTM Layout Aggregation site is a critical failure domain that occurs due to degradation of SFP/XFP modules, fiber cuts, or Rx/Tx power loss. This failure is specific to the Access role on Arista equipment because it relies heavily on the Optical Transport Network (OTN) protocol for transporting customer traffic. The OTN protocol uses the G.709 standard for framing and multiplexing client signals onto the OTU4 (Optical Transport Unit level 4) signal, which is then transmitted over the fiber optic cable. The SFP/XFP modules are responsible for transmitting and receiving this OTU4 signal. Failure of these modules or the fiber optic cable can result in a critical failure of the Access node, leading to service disruptions. In this SOP, we will diagnose and resolve the Optical Transport Failure on BTM-ACC-19 by analyzing the SFP/XFP module status, fiber optic cable integrity, and OTU4 signal quality.

Pre-requisite Checks

- Verify that the BTM Layout Aggregation site is accessible and power is available to the Access node BTM-ACC-19.

- Check the redundant link status to ensure that the primary and secondary links are operational.

- Verify that the dispatch team has been notified and is en route to the site.

- Perform a visual inspection of the fiber optic cable and SFP/XFP modules for any signs of physical damage.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP modules	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces 1/1/1 detail, Expected Output: Port 1/1/1 is up (line protocol is up) Hardware is Ethernet, address is 00:11:22:33:44:55 (bia 00:11:22:33:44:55) MTU 9216 bytes, BW 1000000 Kbit, DLY 10 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Keepalive set (10 sec) Last clearing of 'show interface' counters never Input queue: 0/2000/0/0 (max 2000) Output queue: 0/2000/0/0 (max 2000) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec
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0 packets input, 0 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 0 packets output, 0 bytes, 0 underruns 0 output errors, 0 collisions, 0 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out 0 packets dropped by rate limiter 0 packets output with dribble condition ignored Total Output dropped 0 packets 5 minute output rate 0 bits/sec, 0 packets/sec , Description: This command displays detailed information about the 1/1/1 interface, including its status, MTU, and error counters. The expected output shows that the interface is up and operational.

Command: show optics 1/1/1, Expected Output: Optic 1/1/1 Type: SFP-10G-SR-S Transceiver Model: SFPP-10G-SR-S Vendor: Cisco Serial Number: JHB123456 Rev: A1 Date Code: 200701 Wavelength: 850nm Max Distance: 300m Receive Diagnostics: Temperature: 25°C Optical Power: -2.5dBm Receive Sensitivity: -10.5dBm BER: 1.2e-12 Transmit Diagnostics: Temperature: 25°C Optical Power: 2.5dBm Transmit Sensitivity: -10.5dBm BER: 1.2e-12, Description: This command displays detailed information about the optics of the 1/1/1 interface, including the type, model, and serial number of the SFP module. The expected output shows that the SFP module is a Cisco SFPP-10G-SR-S with a serial number of JHB123456.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optics 1/1/1, Description: This command replaces the SFP/XFP module on the 1/1/1 interface with a new one. The new module will be installed and the old module will be removed.

Step 2: Step: Verify the new SFP/XFP module, Command: show optics 1/1/1, Description: This command verifies that the new SFP/XFP module is installed and operational. The expected output will show that the new module is installed and all diagnostics are within the recommended range.

Post-Check Verification

```
Command: show interfaces 1/1/1 detail, Expected Output: Port 1/1/1 is up (line protocol is up) Hardware is Ethernet, address is 00:11:22:33:44:55 (bia 00:11:22:33:44:55) MTU 9216 bytes, BW 1000000 Kbit, DLY 10 usec, reliability 255/255, txload 1/255, rxload 1/255 Encapsulation ARPA, loopback not set Keepalive set (10 sec) Last clearing of 'show interface' counters never Input queue: 0/2000/0/0 (max 2000) Output queue: 0/2000/0/0 (max 2000) 5 minute input rate 0 bits/sec, 0 packets/sec 5 minute output rate 0 bits/sec, 0 packets/sec 0 packets input, 0 bytes, 0 no buffer Received 0 broadcasts, 0 runts, 0 giants, 0 throttles 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored 0 packets output, 0 bytes, 0 underruns 0 output errors, 0 collisions, 0 interface resets 0 babbles, 0 late collision, 0 deferred 0 lost carrier, 0 no carrier 0 output buffer failures, 0 output buffers swapped out 0 packets dropped by rate limiter 0 packets output with dribble condition ignored Total Output dropped 0 packets 5 minute output rate 0 bits/sec, 0 packets/sec , Description: This command verifies that the 1/1/1 interface is up and operational. The expected output shows that the interface is up and all error counters are zero.

Command: show optics 1/1/1, Expected Output: Optic 1/1/1 Type: SFP-10G-SR-S Transceiver Model: SFPP-10G-SR-S Vendor: Cisco Serial Number: JHB123456 Rev: A1 Date Code: 200701 Wavelength: 850nm Max Distance: 300m Receive Diagnostics: Temperature: 25°C Optical Power: -2.5dBm Receive Sensitivity: -10.5dBm BER: 1.2e-12 Transmit Diagnostics: Temperature: 25°C Optical Power: 2.5dBm Transmit Sensitivity: -10.5dBm BER: 1.2e-12, Description: This command verifies that the new SFP/XFP module is installed and operational. The expected output shows that the new module is installed and all diagnostics are within the recommended range.
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Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: swap optics 1/1/1, Description: This command restores the original SFP/XFP module on the 1/1/1 interface. The original module will be reinstalled and the new module will be removed.